



Electrochemical formation of crystalline $\text{Li}_3\text{VO}_4/\text{Li}_4\text{SiO}_4$ solid solutions film

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ABSTRACT

Crystalline $\text{Li}_3\text{VO}_4/\text{Li}_4\text{SiO}_4$ (LVSO) solid solution films were prepared on Pt substrate electrochemically in absolute ethyl alcohol solution at 150 °C. XRD and Raman structure analyses indicated the presence of characteristic vibrational bands for the orthorhombic structure for the samples. The crystal LVSO thin film has a smooth and dense surface, whose composition was analyzed to be 0.4 (Li_4SiO_4)-0.6(Li_3VO_4). The ionic conductivity of the LVSO film was estimated by impedance spectroscopy to be $8.92 \times 10^{-7} \text{ S/cm}$ at 25 °C with a conductivity activation energy of 0.49 eV.

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1. Introduction

Lithium ion conductors based on the γ - Li_3PO_4 or LISICON structure have been investigated extensively [1,2]; the highest conductivity at room temperature is about $4 \times 10^{-5} \text{ S/cm}$ for γ solid solutions in Li_4GeO_4 - Li_3VO_4 (LVGO) system [3]. Recently Li_4SiO_4 - Li_3VO_4 has been used in rechargeable thin film cells [4–10]. The maximum conductivity was found in the composition around $40\text{Li}_4\text{SiO}_4$ - $60\text{Li}_3\text{VO}_4$ (LVSO), with values of $\sim 1 \times 10^{-5} \text{ S/cm}$ at 20 °C [11]. LVSO has the advantages over LVGO for less expensive to produce, stable in air, no reaction with lithium, low self-discharge, and maintaining its conductivity (constant conductivity) with time. So LVSO is potentially a very attractive material for high temperature battery applications [12].

LVSO powders are generally prepared by sol-gel or solid-state reaction method. And LVSO films are fabricated from LVSO target by techniques such as RF magnetron sputtering or pulsed laser deposition [4–10]. But the as-deposited films show a partial or complete amorphous structure. The amorphous LVSO has somewhat smaller conductivity than the crystalline sample. So some heat treating is necessary for the amorphous thin-film to achieve a crystalline structure [4,5].

In the present work, a new preparation method for LVSO film is presented. Crystalline LVSO thin film could be fabricated in a single step by the method. Electrochemical technique and absolute ethyl alcohol environment are the key points of this method. This new technique has several advantages such as enhanced purity, low reaction temperature. It is economical and environmentally benign over the existing synthetic techniques for LVSO film.

2. Experimental

Platinum metal plates ($20 \times 40 \times 0.2 \text{ mm}^3$, purity 99.98%) were used as the working and counter electrodes. Prior to electrochemical treatment, electrodes were degreased in acetone using an ultrasonic cleaner. Stoichiometric amounts of LiOH , V_2O_5 , and SiO_2 powders were mixed and put in a Teflon beaker together with absolute ethyl alcohol. All the experiments were performed in an autoclave, which was described in detail elsewhere [13]. The Pt substrates were suspended by a Pt wire in the Teflon beaker containing absolute ethyl alcohol in the autoclave. The autoclave was sealed and heated to working temperature of 150 °C. The electrolysis was started with a current density of 0.001 – 1 mA/cm^2 . After each experiment, there was a grey white film appearing on the Pt substrates. The samples were washed with water, ultrasonicated in ethanol, and air-dried prior to characterization.

The crystallographic structure of the samples was analyzed by XRD. The chemical composition of the samples (Li, V, Si) was determined by ICP. Raman spectra were obtained with a microscopic Raman spectrometer using the helium-neon laser line of 633 nm. The cross-sectional view and surface morphology of the deposited specimens were observed by scanning electron microscopy. Ionic conductivity of the LVSO thin film electrolyte was obtained from an AC impedance measurement of Pt/LVSO/Pt sandwich. AC impedance measurements were performed using a CHI660 electrochemical station at frequencies from 0.1 Hz to 100 kHz at different temperature.

3. Result and discussion

Fig. 1 shows the X-ray diffraction patterns of the LVSO film fabricated on a Pt substrate. The as-deposited film indicates diffraction peaks arising from platinum (cubic, Fm3 m) and LVSO, which has an orthorhombic structure with the space group of Pmnb. No impurity

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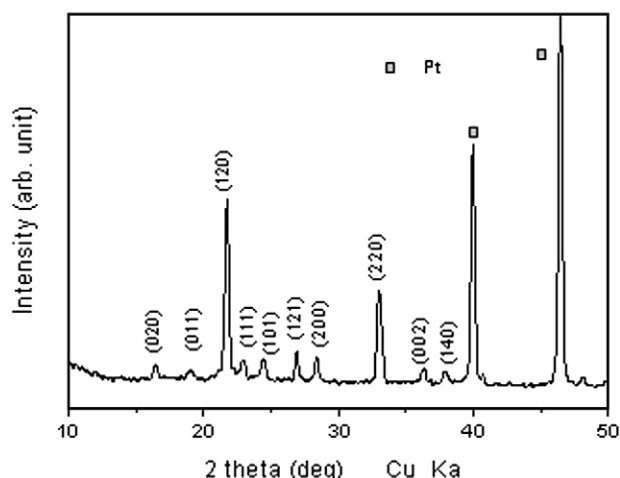


Fig. 1. X-ray diffraction pattern of LVSO film prepared on a Pt.

peaks have been detected. All peaks are sharp, suggesting that the sample is well-crystallized. Use of a low boiling point ethyl alcohol has the potential advantage of optimizing film properties due to increased pressure generated during the reaction [14]. The absolute ethyl alcohol of the reaction system facilitates the formation of the LVSO phase. For the relatively newer technique of utilizing non-aqueous solvents, almost all of the literature deals with powder crystallization. Films grown in organic solvents are finer grained due to the lower solubility of the constituent ions [14–16]. The composition of LVSO films was analyzed by ICP. The chemical composition of the film was $0.4(\text{Li}_4\text{SiO}_4)\cdot 0.6(\text{Li}_3\text{VO}_4)$.

The Raman spectrum of the LVSO film is given in Fig. 2. Bands associating with the stretching and bending vibration of VO_4^{3-} anions have been found to be active in Raman region. The band appearing at about 817 cm^{-1} can be attributed to symmetric stretching of (VO_4^{3-}) , whereas the band at around 787 cm^{-1} can be attributed to asymmetric stretching of (VO_4^{3-}) . Vibration of VO_4 tetrahedra that corresponds to a bending model has been observed at 324 cm^{-1} [17,18]. The strong band of the high-frequency peaks shift to lower wave number comparing with pure Li_3VO_4 (Pmn21) [19]. This indicates that the anion ring has been distorted due to the replacement of V^{5+} ($r_{\text{V}^{5+}}=3.6\text{ nm}$) with Si^{4+} ($r_{\text{Si}^{4+}}=2.6\text{ nm}$). There are no bands in the range from 500 cm^{-1} to 750 cm^{-1} where bands correspond to the valence of vibrations of Si-O-Si bridge [20]. It indicates that there is no Li_2SiO_3 impurity in the sample.

The scanning electron micrograph of the oxide film formed on platinum substrates is shown in Fig. 3. Thin film is consisted of

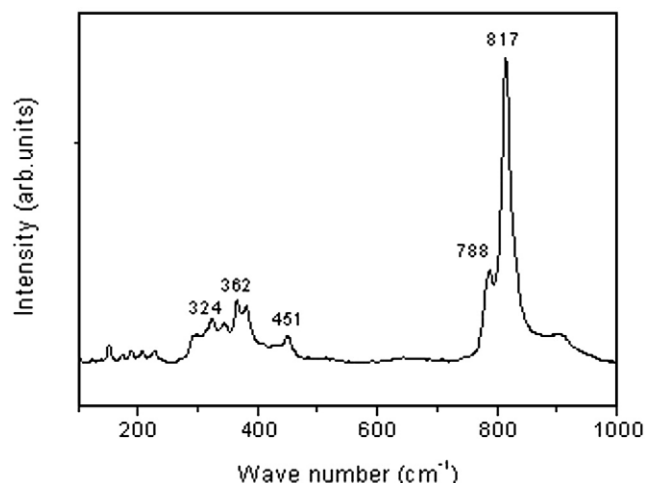


Fig. 2. Raman spectrum for LVSO.

homogenous spherical particles, the diameters of the spheres are around 100 nm . The film is smooth, dense and free of pinholes and cracks. The characterization shows an excellent quality as a solid electrolyte separating cathode and anode materials. The cross-sectional view shown in Fig. 3 (c) demonstrates that the thin film is well adherent to the substrate surface with a uniform thickness. The thickness is estimated to be $8.47\text{ }\mu\text{m}$.

Electrical conductivity of the LVSO thin film is shown in Fig. 4. The impedance spectra exhibit the typical behavior of a single dielectric relaxation process for ionic conductor between two blocking electrodes. The electrolyte film resistance can be obtained by selecting the value of real part $\text{Re}(Z)$ at the frequency where the imaginary part $-\text{Im}(Z)$ goes through a local minimum, which is the same as the value obtained from a complex impedance plot. The ionic conductivity of

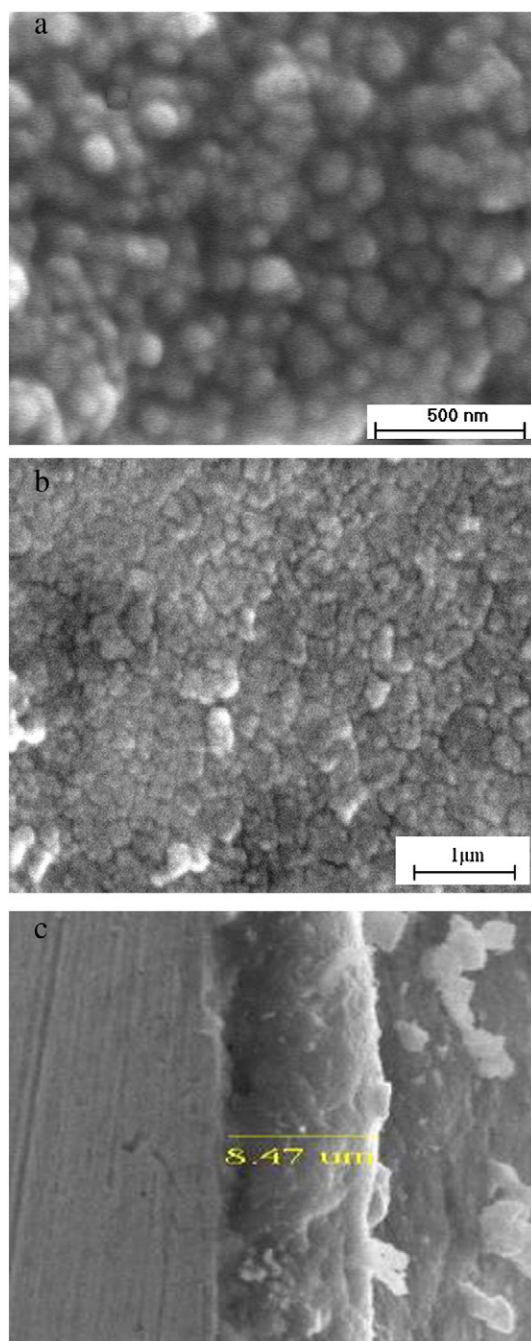


Fig. 3. Oxide film formed on Pt substrate.

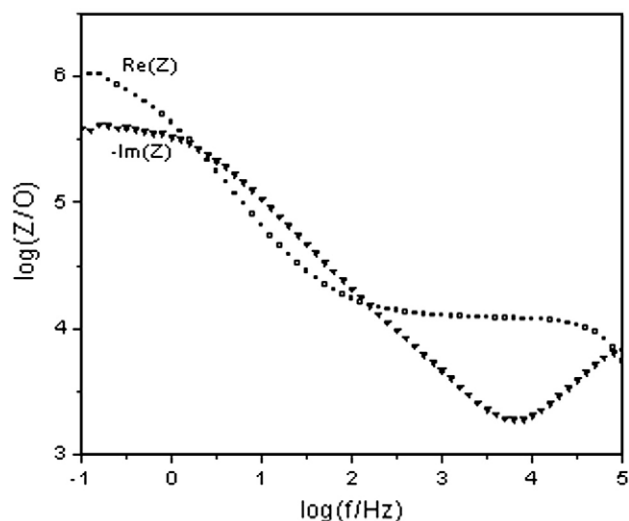


Fig. 4. AC impedance spectra for LVSO thin film.

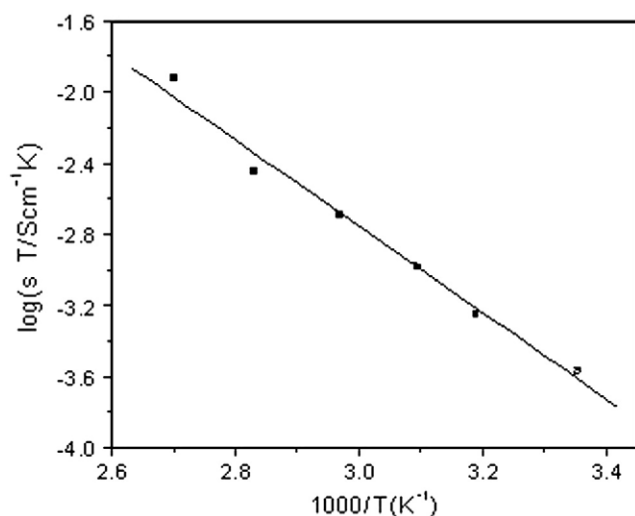


Fig. 5. The plot of $\log(\sigma T)$ vs. $1000/T$ for LVSO.

the LVSO film is estimated from the resistances and the thickness of the film. The conductivity at 25 °C is about 8.92×10^{-7} S/cm, which is higher than the value of $1\text{--}3.98 \times 10^{-7}$ S/cm reported for the film prepared by PLD [5–7]. The difference is probably due to the crystallized form of the present film and the complete amorphous nature of the PLD film. The RF-sputtered film with the partially

crystallized form showed ca. 1.0×10^{-6} S/cm at room temperature [8–10], which is a little higher than the present thin film. This is because the former is more compact than the latter. From the temperature dependence of the ionic conductivity shown in Fig. 5, the activation energy is estimated to be 0.49 eV.

4. Conclusion

Crystal LVSO thin film can be produced electrochemically by oxidation of silicate, vanadium and lithium species present in absolute ethyl alcohol solution. XRD and Raman structure analyses indicate the presence of characteristic vibrational bands for the orthorhombic structure for the samples. This synthetic method can form a crystalline LVSO thin film in a single step. A dense, smooth and uniformly thick Li–V–Si–O thin film without pinholes and cracks was obtained at 150 °C. In this condition, the prepared thin film exhibits an ionic conductivity of 8.92×10^{-7} S/cm at 25 °C with conductivity activation energy of 0.49 eV. The electrochemical synthesis in absolute ethyl alcohol would facilitate applications of other orthosilicates or other materials.

Acknowledgements

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